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Clinical Sports Medicine Update

Platelet-Rich Plasma Injection for the Treatment of Hamstring Injuries

A Systematic Review and Meta-analysis With Best-Worst Case Analysis

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Background: Hamstring injuries are common and account for considerable time lost to play in athletes. Platelet-rich plasma has potential as a means to accelerate healing of these injuries.

Purpose: (1) To present the evidence of platelet-rich plasma injection in the treatment of hamstring injuries, (2) evaluate the ''best-case scenario'' in dichotomous outcomes, and (3) evaluate the ''worst-case scenario'' in dichotomous outcomes.

Study Design: Systematic review and meta-analysis.

Methods: Two authors systematically reviewed the PubMed, Embase, and Cochrane Library databases according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, with any discrepancies resolved by mutual consensus. The level of evidence was assessed per the criteria of the Oxford Centre for Evidence-Based Medicine and the quality of evidence by the Coleman Methodology Score. Meta-analysis by fixed effects models was used if heterogeneity was low ($I^2 < 25\%$) and random effects models if heterogeneity was moderate to high ($I^2 \geq 25\%$). P values $<.05$ were considered statistically significant.

Results: A total of 10 studies were included with 207 hamstring injuries in the platelet-rich plasma group and 149 in the control group. The fixed effects model for mean time to return to play, which compared platelet-rich plasma + physical therapy and physical therapy alone, significantly favored platelet-rich plasma + physical therapy (mean difference, –5.70 days). The fixed effects model for reinjury rates, which also compared platelet-rich plasma + physical therapy with physical therapy alone insignificantly favored platelet-rich plasma + physical therapy (risk ratio, 0.88). The best-case scenario fixed effects model for reinjury rates insignificantly favored platelet-rich plasma + physical therapy (risk ratio, 0.82). The worst-case scenario fixed effects model for reinjury rates insignificantly favored physical therapy alone (risk ratio, 1.13). The mean ± SD complication rate for either post-injection discomfort, pain, or sciatic nerve irritation was 5.2% ± 2.9% (range, 2.7% to 9.1%).

Conclusion: There has been statistically nonsignificant evidence to suggest that PRP injection ± PT reduced mean time to RTP or reinjury rates compared to no treatment or PT alone for hamstring injuries in a short-term follow-up. The complication profiles were favorable. Further studies of high quality and large cohorts are needed to better support or disprove the consensus of the systematic review and meta-analysis.

Keywords: platelet-rich fibrin; reinjury rates; return to play

Hamstring injuries are common in all athletes but particularly those involved in high-speed running, acceleration, deceleration, and rapid directional changes.[9,22,41] These injuries can lead to considerable time lost to play.[22] The event of chronic cicatrization of the muscle and/or tendon, which leads to chronic myotendinopathy, has been an area of concern and therefore has added importance for the identification of more efficacious treatments, whether adjuvant or alternative, to the nonsurgical and surgical management of hamstring injuries.[42] This has led to an increasing number of research studies to investigate techniques to augment the healing process of myotendinous injuries to expedite return to play (RTP).[2,12,20,30,38,42]

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Platelet-rich plasma (PRP) offers the prospect to augment the healing response through the intrinsic myotendinous inductive properties of growth factors in concentrated autologous blood.[4,28] In vitro studies have demonstrated these effects to be the stimulation of cellular migration, cellular proliferation, angiogenesis, and matrix deposition.[5,19] The minimally invasive nature of the treatment technique, coupled with its theoretically sound mechanism of action, provides a rationale for clinical investigation.

Numerous studies have reported the success of PRP injection for various tendinopathies and muscle injuries.[15,23,37,43,51] However, no systematic review to date has clarified the efficacy of PRP injections in the treatment of hamstring injuries alone. Furthermore, no study has provided a "best- and worst-case scenario" meta-analysis of dichotomous outcomes, where all patients lost to follow-up were adjusted to either not experience or experience the negative outcome specific to that analysis, respectively. This is an analytical method that can help to negate selection bias inherent in studies with patient loss to follow-up and assessment bias inherent in retrospective studies.

The purpose of this study was to (1) present the evidence of PRP injection for hamstring injuries, (2) evaluate the best-case scenario in dichotomous outcomes, and (3) evaluate the worst-case scenario in dichotomous outcomes.

METHODS

Study Design, Search Strategy, and Study Selection

Two authors (D.S. and T.N.B.T.Y.) systematically reviewed the PubMed, Embase, and Cochrane Library databases from their inception to August 2018 according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines.[40] The search terms used were as follows: (hamstring OR hamstring strain OR hamstring tear OR hamstring injury OR hamstring repair OR arthroscopic hamstring repair OR hamstring arthroscopy) AND (PRP OR platelet OR platelet-rich OR platelet-rich-plasma OR platelet-rich-fibrin). The inclusion criteria were clinical studies that reported outcomes after PRP injections for hamstring injuries and were full-text articles written in English and published in a peer-reviewed journal. The exclusion criteria were reviews, cadaveric studies, in vitro studies, animal studies, and case reports. The titles, abstracts, and full texts of studies identified through the search terms were then screened according to the aforementioned inclusion and exclusion criteria. Studies were included according to the concordance of 2 authors (D.S. and T.N.B.T.Y.), with any disagreements resolved by consultation of a third author (Y.S.).

Assessment of Evidence

Level of evidence was based on the Oxford Centre for Evidence-Based Medicine.[32] The quality of evidence was evaluated with the Coleman Methodology Score, which assesses the methodological quality of a study's design, with scores between 0 and 100.[13] Studies were considered to be of excellent quality from 85 to 100 points, good from 70 to 84, fair from 55 to 69 points, and poor if <55.[34] A maximum score of 100 represents a study design that is largely devoid of the influence of chance, differential biases, and confounding factors. The overall quality of the literature was further appraised with the grades of recommendation by *The Journal of Bone and Joint Surgery*.[56] Grade A indicates good quality of literature composed of level 1 studies with consistent findings; grade B, fair quality composed of level 2 or 3 studies with consistent findings; grade C, poor quality with level 4 or 5 studies, not allowing a recommendation; and grade I, insufficient evidence to provide a recommendation.

Data Extraction

The data of the included studies were extracted onto a standardized data sheet with the following variables analyzed: functional outcome scores, mean time to RTP, reinjury rates, and complications. The following characteristics of PRP injections were also extracted and presented: number of injections, volume, preparation kit, leukocyte concentration, platelet concentration, activating agent, and anticoagulant used.

Definition of ''As Protocol'' and ''Per Protocol''

As protocol: the total number of patients recruited at the start of the study, with loss to follow-up not accounted for yet.
Per protocol: the total number of patients after treatment and follow-up, with loss to follow-up accounted for.[46]

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Statistical Analysis

Statistical analysis was performed with RevMan (v 5.3; The Nordic Cochrane Centre). Results were meta-analyzed if ≥3 studies reported an outcome.[33] Heterogeneity among studies was quantified with the I^2 statistic, with values <25% considered low heterogeneity, 25% to 50% moderate, 51% to 75% moderate to high, and >75% high.[31] Continuous outcomes were expressed as the mean difference and dichotomous outcomes as the risk ratio. Fixed effects models were used if heterogeneity was low, and random effects models were used if heterogeneity was moderate to high.[27] If the median and range/interquartile range were provided, previously published methods by Wan et al[54] were used to estimate the mean and standard deviation, respectively. P values <.05 were considered statistically significant.

RESULTS

Literature Search

A literature search based on the search strategy yielded 1147 studies for review, with 10 studies included (Figure 1). An additional study was identified as having been published following the literature search date, during data extraction.[39] The included studies were published from 2013 to 2018.[#]

Study Characteristics and Patient Demographics

The mean ± SD level of evidence was 2.9 ± 0.9 (range, 2-4), and the mean quality of evidence was 70.5 ± 10.3 (range, 61-88) (Table 1). Four studies were regarded as having good to excellent methodological quality.[1,16,29,48] All hamstring injuries were confirmed by either ultrasound or magnetic resonance imaging. For hamstring injuries treated as protocol, 207 hamstrings were in the PRP group and 149 in the control group. These included 69 grade I hamstring injuries, 191 grade II, 6 grade III, and 90 of unclassified severity. For hamstrings treated per protocol, 190 were in the PRP group and 141 in the control group. Mean follow-up was 9.1 ± 10.7 months (range, 2-36.6 months).

PRP Injection Characteristics

Variations in the characteristics of PRP injections were present across the included studies (Table 2). Eight studies guided PRP injection by ultrasound,[1,16,21,26,39,47,48,59] and 2 had blinded injection.[29,55] The number of injections administered were predominantly single, with 2 studies employing 2 or 3 total injections.[48,59] One of these studies employed 2 or 3 injections per the grade of hamstring injury,[59] whereas the other study employed 2 injections for all hamstring injuries as protocol.[48] The volume of PRP administered ranged from 2.5 to 6 mL. The most common preparation kit employed was the Biomet GPS-III system.[1,29,47,55] Of the 5 studies with leukocyte concentrations

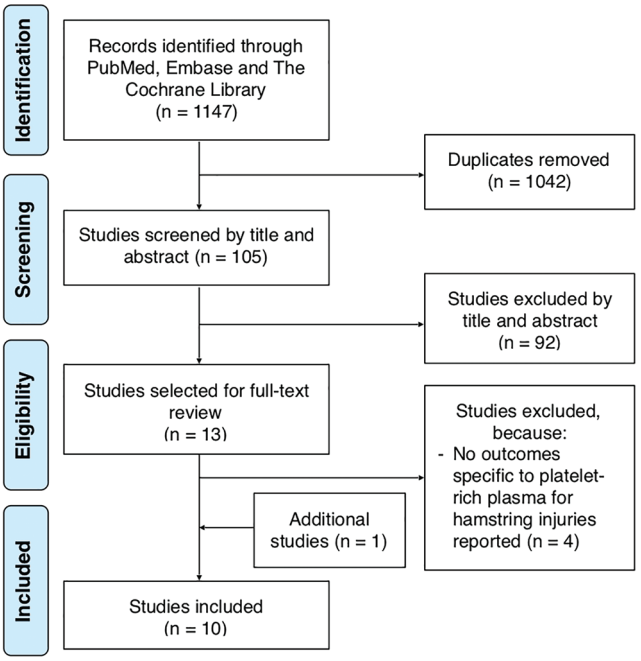

Figure 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram.

indicated, 3 studies used leukocyte-rich PRP injections,[1,16,29] and 2 had leukocyte-poor PRP injections.[21,47] The mean platelet concentration was 851.5 ± 357.9 × 10^3 μL (range, 433-1297).[1,29,39,48] Local anesthesia of 1% lidocaine was used in 3 studies.[16,21,39]

Physical Therapy Characteristics

Physical therapy (PT) was employed either in addition to PRP injections or as a comparative treatment arm in 8 of the 10 studies.[1,21,26,29,47,48,55,59] No description of PT was present in 4 of these studies.[1,21,55,59] A lack of a standardized PT regimen was observed among the remaining 4 studies, with a variety of regimens described.[26,29,47,48]

Functional Outcomes

In the included 10 studies, 6 reported functional outcomes by 5 scoring systems.[1,21,29,39,48,55] Among these scoring systems, visual analog scale for pain was the most common, as employed in 4 studies.[21,29,48,55]

There was a notable lack of complete reporting for functional outcomes from baseline to follow-up, with only 2 of the 6 studies doing so.[39,55] In these studies, significant improvement was observed in the visual analog scale and Nirschl Pain Rating Scale,[55] whereas nonsignificant improvement was observed in the Victoria Institute of Sports Assessment–Hamstring score.[39]

Time to RTP

The mean time to RTP in the PRP group ranged from 21.0 to 50.9 days compared to the control group that ranged

[#]References 1, 16, 21, 26, 29, 39, 47, 48, 55, 59.

TABLE 1
Characteristics of Included Studies[a]

Study	LOE	QOE	Treatment	Hamstrings as Protocol, No.	Hamstrings per Protocol, No.	Classification as Protocol	Mean Follow-up (Range), mo	Functional Outcome Scores, Baseline → Follow-up	Mean Time to RTP, d	Reinjury Rates, %	Complications
Wetzel[55]	3	61	PRP + PT, PT	12, 5	NR	NR	3.25 (1-24)	VAS: 8.2 → 0.7; 7.4 → 1.2 NPRS: 5.5 → 1.5; 4.4 → 2.0	NR	NR	None
Rettig[47]	3	66	PRP, PT	5, 5	NR	4 grade II, 6 grade III	6	NR	21.5, 30.8	0, 0	None
A Hamid[1]	2	79	PRP, PT	14, 14	12, 12	28 grade II	NR	BPI-SF: 3.9 → NR; 4.3 → NR	26.7, 42.5	NR	None
Davenport[16]	2	74	PRP + PT, whole blood injection	12, 7	11, 6	NR	6	NR	NR	NR	1 sciatic nerve irritation
Fader[21]	4	61	PRP	18	NR	NR	6	VAS: 4.6 → NR	NR	NR	1 postinjection discomfort
Guillodo[26]	3	66	PRP + PT, PT	15, 19	NR	34 grade II	4	NR	50.9, 52.8	NR	None
Hamilton[29]	2	88	PRP + PT, PPP + PT, PT	30, 30, 30	26, 28, 29	46 grade I, 43 grade II, 1 unclassified	6	VAS: 6.8 → NR; 6.6 → NR; 6.9 → NR	medians: 21, 27, 25	7.7, 10.7, 10.3	None
Reurink[48]	2	85	PRP + PT, placebo injection	41, 39	37, 37	23 grade I, 57 grade II	12	VAS: NR → 0.2; NR → 0.1 HOS: NR → 86; NR → 88	43.3, 45	27, 30	1 postinjection pain
Zanon[59]	4	64	PRP + PT	25	NR	25 grade II	36.6 (22-42)	NR	36.8	12	None
Levy[39]	4	61	PRP	35	29	NR	2	VISA-H: 43.9 → 51.14	NR	NR	1 postinjection pain

[a]BPI-SF, Brief Pain Inventory–Short Form; HOS, Hamstring Outcome Score; LOE, level of evidence; NPRS, Nirschl Pain Rating Scale; NR, not reported; PPP, platelet-poor plasma; PRP, platelet-rich plasma; PT, physical therapy; QOE, quality of evidence; RTP, return to play; VAS, visual analog scale; VISA-H, Victoria Institute of Sports Assessment–Hamstring.

from 30.8 to 52.8 days. Mean time to RTP was not reported in 4 of 10 studies[16,21,39,55], with this accounted for 89 of the 356 hamstrings as protocol. Based on the random-effects model for time to RTP that compared PRP + PT to PT alone, the mean difference was nonsignificantly -5.67 (95% CI, -12.62 to 1.28; I^2 = 36%; P = .11) days in favor of PRP + PT (Figure 2).

Reinjury Rates

No specific definition of reinjury was provided, with the confirmation of reinjury by any imaging modality not described. The mean reinjury rate in the PRP group was 11.7% ± 11.4% (range, 0%-27%) as compared with the control group at 12.8% ± 12.5% (range, 0%-30%). Based on the fixed effects model for reinjury rates that compared PRP + PT with PT alone, the risk ratio was 0.88 (95% CI, 0.45-1.71; I^2 = 0%; P = .70) in favor of PRP + PT (Figure 3). In the "best-case scenario" fixed effects model for reinjury rates, which assumed that all patients lost to follow-up did not experience reinjury, the risk ratio was 0.82 (95% CI, 0.42-1.62; I^2 = 0%; P = .57) in favor of PRP + PT (Figure 4). In the "worst-case scenario" fixed effects model for reinjury rates, which assumed that all patients lost to follow-up experienced reinjury, the risk ratio was 1.13 (95% CI, 0.66-1.96; I^2 = 0%; P = .65) in favor of PT alone (Figure 5). No statistical differences were identified in reinjury rates between PRP + PT and PT alone. Mean time to reinjury was not reported.

Complications

No complications were indicated in 6 studies,[1,26,29,47,55,59] and a single complication each was noted in the remaining 4 studies. This totaled 4 complications, with a mean complication rate of 5.2% ± 2.9% (range, 2.7%-9.1%) of postinjection discomfort, pain, or sciatic nerve irritation.[16,21,39,48] No infections were reported. Follow-up of the sciatic nerve irritation by ultrasound revealed no abnormalities.[16] Two of these complications resolved spontaneously,[21,39]

TABLE 2
Characteristics of Platelet-Rich Plasma Injection[a]

Study	Technique	No. of Injections	Volume, mL	Preparation Kit	Concentration, $\times 10^3$, μL: Leukocyte	Concentration, $\times 10^3$, μL: Platelet	Activating Agent	Anticoagulant	Local Anesthetic
Wetzel[55]	Blinded	1[b]	6	Biomet GPS-III system (Biomet)	NR	NR	NR	NR	NR
Rettig[47]	Ultrasound guided	1	NR	Biomet GPS-III system (Biomet)	NR	NR	NR	NR	NR
A Hamid[1]	Ultrasound guided	1	3	Biomet GPS-III system (Biomet)	Leukocyte rich, 38.3	1297	None	NR	NR
Davenport[16]	Ultrasound guided	1	3	Smart PReP (Harvest Technologies)	Leukocyte rich	NR	None	NR	1% lidocaine
Fader[21]	Ultrasound guided	1	2.5-4	Angel Whole Blood Separation System (Cytomedix Inc)	Leukocyte poor	NR	NR	NR	2-5 mL of 1% lidocaine with 10% bicarbonate dilution
Guillodo[26]	Ultrasound guided	1	3	Pras 20 kit (Proteal)	NR	NR	NR	NR	NR
Hamilton[29]	Blinded	1	3	Biomet Recover, GPS-III Platelet Separation System (Biomet)	Leukocyte rich, 26.1	765.8	None	Citrate dextrose solution	NR
Reurink[48]	Ultrasound guided	2[c]	3	Arthrex ACP double-syringe system (Arthrex Medizinische Instrumente GmbH)	Leukocyte poor, 1.9	433	NR	NR	NR
Zanon[59]	Ultrasound guided	2, grade 2a; 3, grade >2a[d]	2.5-3	NR	NR	NR	Calcium gluconate	Dipotassium EDTA	NR
Levy[39]	Ultrasound guided	1	6	RegenKit BCT-3 (Regenlab)	NR	910	NR	NR	4 mL of 1% lidocaine

[a]EDTA, ethylenediaminetetraacetic acid; NR, not reported.
[b]One patient had second injection at 6.5 months.
[c]Five- to 7-day interval.
[d]One-week interval.

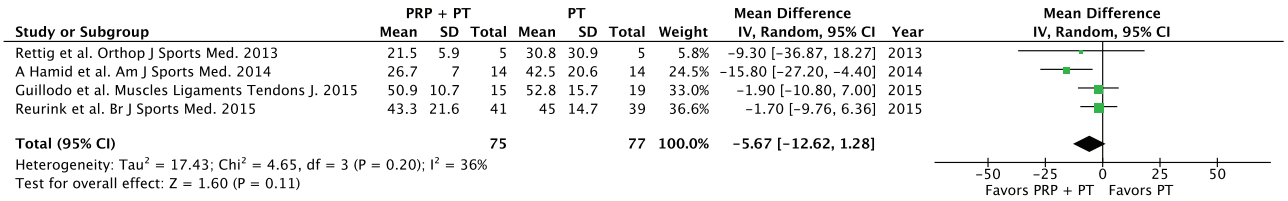

Figure 2. Mean-analysis of time to return to play, random-effects model. IV, inverse variance; PRP, platelet-rich plasma; PT, physical therapy.

whereas the prognosis of the remaining 2 complications was not reported.[16,48]

DISCUSSION

The high incidence of hamstring injuries in athletes has prompted investigation into treatment innovations that reduce recovery time (time to RTP) and decrease reinjury rates. PRP has been reported to have therapeutic effects for various myotendinous injuries, but its evidence has been limited when used to treat hamstring injuries.[5,15,23,25,43,44] In vitro studies have indicated that PRP can induce healing by the reduction of inflammation and the upregulation of an anabolic response through the release of transforming growth factor β, platelet-derived endothelial growth factor, insulin-like growth factor 1, hepatocyte growth factor, vascular endothelial growth factor, and fibroblast growth factors.[5,7,14,44] Therefore, the findings of this meta-analysis are surprising given the

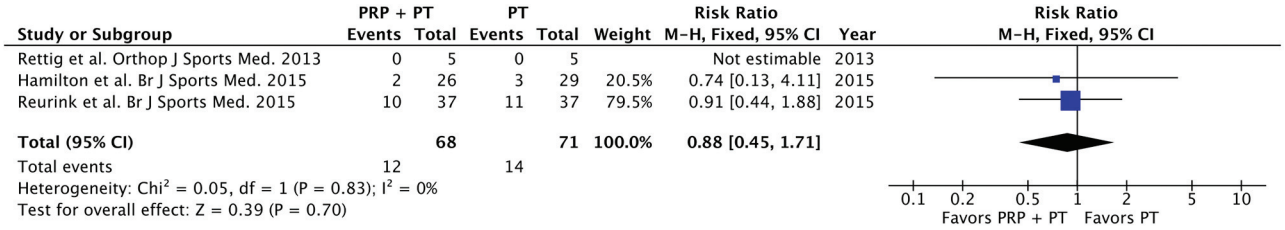

Figure 3. Fixed effects model: meta-analysis of reinjury rates as reported. M-H, Mantel-Haenszel; PRP, platelet-rich plasma; PT, physical therapy.

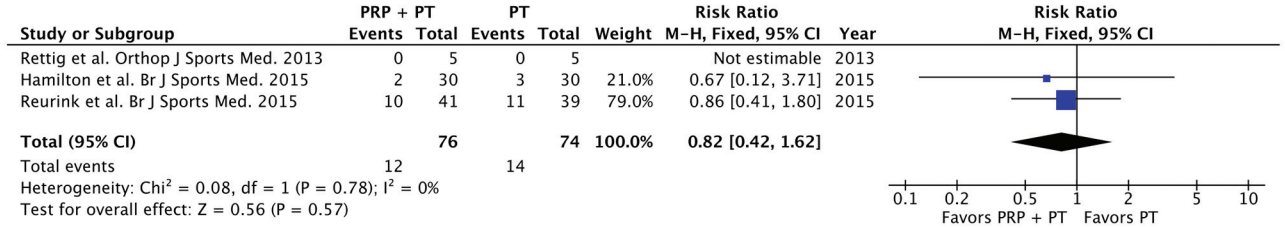

Figure 4. Fixed effects model: ''best-case scenario'' meta-analysis of reinjury rates. M-H, Mantel-Haenszel; PRP, platelet-rich plasma; PT, physical therapy.

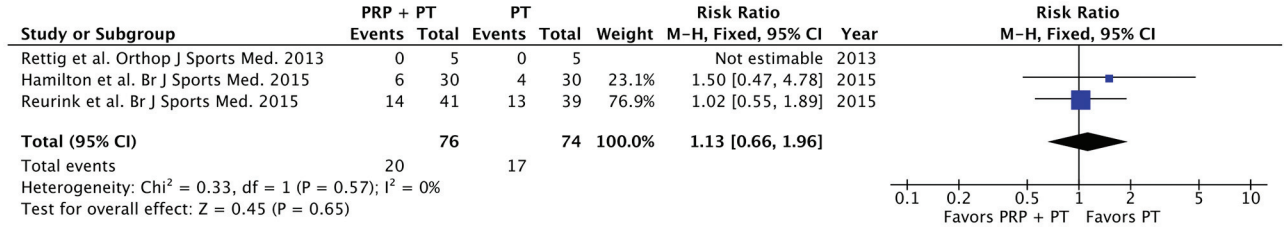

Figure 5. Fixed effects model: ''worst-case scenario'' meta-analysis of reinjury rates. M-H, Mantel-Haenszel; PRP, platelet-rich plasma; PT, physical therapy.

basic science evidence supporting PRP in the augmentation of soft tissue healing.[15,23,37,43,51] However, this meta-analysis demonstrated that PRP appeared not to have an efficacious effect on reducing reinjury rates. This is interesting, as in vitro studies have demonstrated that PRP can improve the quality of the tissue repair.[60] It is arguable that circumstances that surround hamstring reinjuries can encompass factors beyond medical control but instead be related to the nature of the sport itself.

Despite the common application of PRP injections, reports on the efficacy of PRP for soft tissue injuries have been inconsistent in clinical trials largely because of the lack of a standardized protocol for PRP preparation and injection.[49] This meta-analysis was limited by the available evidence to perform subgroup meta-analysis by individual PRP injection characteristics. Notably, this meta-analysis has highlighted that no standardized protocol exists for the timing of the PRP injection and the number of injections required. This confounds meaningful analysis. While basic science evidence suggests that the timing of a PRP therapy should be in the postinflammatory phase, no such evidence exists in any clinical study on soft tissue injury, let alone for hamstring injuries specifically.[11]

The most commonly employed PRP preparation kit in this meta-analysis, the Biomet GPS-III system, has been associated with PRP preparations of high leukocyte concentrations and a low-density fibrin network after activation.[19] PRP preparations containing a high leukocyte concentration have been reported to produce greater local inflammatory responses and therefore potentiate postinjection pain.[57] Increased cell death and synoviocyte activation has been postulated to occur at a greater degree with leukocyte-rich PRP as compared with leukocyte-poor PRP.[8] Leukocyte-rich PRP has also been implicated in expressing higher levels of collagen type III and, as such, may contribute to scar formation.[61] Higher concentrations of neutrophils in PRP have also been seen to express metalloproteases (MMP-8 and MMP-9) and reactive oxygen species that can catabolize cells and reduce soft tissue healing.[3]

This evidence remains inconclusive as leukocyte-rich PRP has demonstrated the potential to break down scar tissue in chronic Achilles tendinopathy in the rat model,

which may augment the healing process.[36] Yoshida and Murray[58] demonstrated in an in vitro study that fibroblasts cultured with platelets and only peripheral blood mononuclear cells (no neutrophils present) increased the cell metabolic activity desired for healing by insulin-like growth factor 6 expression. It was interesting to note that these factors were not considered or not available for consideration, as the majority of studies did not report the leukocyte concentrations or the specific components in leukocyte concentrations of the injected PRP. Greater detail in reporting of PRP injection characteristics is warranted to allow future studies to investigate the optimal PRP injection protocol.

The addition of anticoagulants during PRP preparations has been an area of inconsistency. Citrate compounds and EDTA (ethylenediaminetetraacetic acid) were the preparations employed in the included studies.[29,59] The use of anticoagulation aims to increase platelet concentration levels and therefore enable greater therapeutic effects. In vitro studies have indicated that EDTA yielded higher platelet concentrations than heparin and that heparin yielded higher platelet concentrations than citrate compounds.[18,53] To achieve greater therapeutic effects, platelet-activating agents have similarly been used. This aims to degranulate platelets and cleave fibrinogen to initiate prompt growth factor release to aid healing.[10] Nguyen et al[45] indicated that calcium chloride or thrombin can prompt the release of nearly 70% of the growth factors contained within the platelets in PRP. However, DeLong et al[17] reported that early and rapid activation of the PRP can be disadvantageous, as there may be a decrease in total amount of growth factors available at the injury site over time. Studies with more detailed reporting of PRP injection characteristics are warranted, before conclusions can be made.

The use of analgesics to reduce pain before PRP injections has also been an area of inconsistent reporting. Analgesics have been indicated to negatively affect platelet function and prevent the release of bioactive compounds crucial for healing.[6,50] Bausset et al[6] reported that local analgesics, such as lidocaine and ropivacaine, can inhibit platelet aggregation but may not affect the capacity of the platelets to release growth factors. Schippinger et al[50] indicated that the intake of nonsteroidal anti-inflammatory drugs may affect the efficacy of PRP, as the inhibition of cyclooxygenase reduces the platelet activation and aggregation required for healing. Therefore, the reported use of analgesics to ease pain before PRP injections for hamstring injuries must be employed with caution, as the "standard of care" may underscore the need for "best practice."

This systematic review and meta-analysis had several limitations. Of the 10 included studies, 5 contained >2 patients lost to follow-up, and the remaining 5 studies failed to state if any patients were lost to follow-up. This predisposed selection bias that may have overestimated the therapeutic potential of PRP for hamstring injuries. The majority of studies failed to include a sham injection arm to control against the placebo effect. The included studies included varying degrees in severity of hamstring injuries. As the prognoses for varying hamstring grades are distinct, the generalization of outcomes within the study and then the meta-analysis must be appreciated with this limitation in view. Outcomes broken down by degree of hamstring injury were not possible owing to the limited reporting standards. The literature search was limited to the PubMed, Embase, and Cochrane Library databases with only full-text studies published in English and therefore may have succumbed to selection bias. The inclusion of retrospective studies in the meta-analysis added selection and assessor bias, as the inherent design of these studies can overenthusiastically report outcomes. Protocols for PRP injection and PT lacked standardization in the literature and may have thus added ambiguity to the results reported. The poor reporting of PRP injection characteristics limited subgroup analysis and confounded meaningful analysis.

Meta-analysis by best- and worst-case scenario for reinjury rates was performed to aid negation of selection biases. This analytical method provides an absolute confidence interval to reflect the uncertainty of missing binary data.[24] Patients lost to follow-up may not necessarily have poorer results; therefore, best-case scenario analysis can be regarded as being equally important compared with worse-case scenario analysis.[35] Clinicians may subsequently better recognize by overview the extreme optimistic (best-case scenario analysis) and pessimistic (worse-case scenario analysis) outcome possibilities of PRP for hamstring injuries. This is important information that can aid clinicians to better consult patients in their treatment decision.[52]

CONCLUSION

The systematic review has demonstrated that there has been statistically nonsignificant evidence to suggest that PRP injection ± PT reduced time to RTP or re-injury rates compared to no treatment or PT alone for hamstring injuries in a short-term follow-up. The favorable complication profiles observed underscore the safety of treatment studied. Heterogeneity of PRP injection protocols should be controlled and accounted for in future studies. Variables accounted for in Table 2 can be used as a reporting guideline standard for the characteristics of PRP injections. Standardization of PRP injection protocols is warranted with emphasis on the timing and number of PRP injections specific to hamstring injury severity. Until such standardization of intervention is controlled and established, only a C grade recommendation can be advised for the use of PRP in the treatment of hamstring injuries based on the available evidence.[56]

REFERENCES

1. A Hamid MS, Mohamed Ali MR, Yusof A, George J, Lee LP. Platelet-rich plasma injections for the treatment of hamstring injuries: a randomized controlled trial. *Am J Sports Med*. 2014;42(10):2410-2418.
2. Almekinders LC. Anti-inflammatory treatment of muscular injuries in sport: an update of recent studies. *Sports Med*. 1999;28(6):383-388.
3. Anitua E, Sánchez M, Nurden AT, Nurden P, Orive G, Andía I. New insights into and novel applications for platelet-rich fibrin therapies. *Trends Biotechnol*. 2006;24(5):227-234.

4. Arnoczky SP, Sheibani-Rad S. The basic science of platelet-rich plasma (PRP): what clinicians need to know. *Sports Med Arthrosc Rev*. 2013;21(4):180-185.
5. Baksh N, Hannon CP, Murawski CD, Smyth NA, Kennedy JG. Platelet-rich plasma in tendon models: a systematic review of basic science literature. *Arthroscopy*. 2013;29(3):596-607.
6. Bausset O, Magalon J, Giraudo L, et al. Impact of local anaesthetics and needle calibres used for painless PRP injections on platelet functionality. *Muscles Ligaments Tendons J*. 2014;4(1):18-23.
7. Beck J, Evans D, Tonino PM, Yong S, Callaci JJ. The biomechanical and histologic effects of platelet-rich plasma on rat rotator cuff repairs. *Am J Sports Med*. 2012;40(9):2037-2044.
8. Braun HJ, Kim HJ, Chu CR, Dragoo JL. The effect of platelet-rich plasma formulations and blood products on human synoviocytes: implications for intra-articular injury and therapy. *Am J Sports Med*. 2014;42(5):1204-1210.
9. Brooks JH, Fuller CW, Kemp SP, Reddin DB. Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34(8):1297-1306.
10. Cavallo C, Roffi A, Grigolo B, et al. Platelet-rich plasma: the choice of activation method affects the release of bioactive molecules. *Biomed Res Int*. 2016;2016:6591717.
11. Chan BP, Fu SC, Qin L, Rolf C, Chan KM. Supplementation-time dependence of growth factors in promoting tendon healing. *Clin Orthop Relat Res*. 2006;448:240-247.
12. Chu SK, Rho ME. Hamstring injuries in the athlete: diagnosis, treatment, and return to play. *Curr Sports Med Rep*. 2016;15(3):184-190.
13. Coleman BD, Khan KM, Maffulli N, Cook JL, Wark JD; Victorian Institute of Sport Tendon Study Group. Studies of surgical outcome after patellar tendinopathy: clinical significance of methodological deficiencies and guidelines for future studies. *Scand J Med Sci Sports*. 2000;10(1):2-11.
14. Cross JA, Cole BJ, Spatny KP, et al. Leukocyte-reduced platelet-rich plasma normalizes matrix metabolism in torn human rotator cuff tendons. *Am J Sports Med*. 2015;43(12):2898-2906.
15. Dallaudière B, Pesquer L, Meyer P, et al. Intratendinous injection of platelet-rich plasma under US guidance to treat tendinopathy: a long-term pilot study. *J Vasc Interv Radiol*. 2014;25(5):717-723.
16. Davenport KL, Campos JS, Nguyen J, Saboeiro G, Adler RS, Moley PJ. Ultrasound-guided intratendinous injections with platelet-rich plasma or autologous whole blood for treatment of proximal hamstring tendinopathy: a double-blind randomized controlled trial. *J Ultrasound Med*. 2015;34(8):1455-1463.
17. DeLong JM, Russell RP, Mazzocca AD. Platelet-rich plasma: the PAW classification system. *Arthroscopy*. 2012;28(7):998-1009.
18. do Amaral RJ, da Silva NP, Haddad NF, et al. Platelet-rich plasma obtained with different anticoagulants and their effect on platelet numbers and mesenchymal stromal cells behavior in vitro. *Stem Cells Int*. 2016;2016:7414036.
19. Dohan Ehrenfest DM, Rasmusson L, Albrektsson T. Classification of platelet concentrates: from pure platelet-rich plasma (P-PRP) to leucocyte- and platelet-rich fibrin (L-PRF). *Trends Biotechnol*. 2009;27(3):158-167.
20. Ekstrand J, Hägglund M, Waldén M. Injury incidence and injury patterns in professional football: the UEFA injury study. *Br J Sports Med*. 2011;45(7):553-558.
21. Fader RR, Mitchell JJ, Traub S, et al. Platelet-rich plasma treatment improves outcomes for chronic proximal hamstring injuries in an athletic population. *Muscles Ligaments Tendons J*. 2015;4(4):461-466.
22. Feeley BT, Kennelly S, Barnes RP, et al. Epidemiology of National Football League training camp injuries from 1998 to 2007. *Am J Sports Med*. 2008;36(8):1597-1603.
23. Filardo G, Kon E, Della Villa S, Vincentelli F, Fornasari PM, Marcacci M. Use of platelet-rich plasma for the treatment of refractory jumper's knee. *Int Orthop*. 2010;34(6):909-915.
24. Gamble C, Hollis S. Uncertainty method improved on best-worst case analysis in a binary meta-analysis. *J Clin Epidemiol*. 2005;58(6):579-588.
25. Gaweda K, Tarczynska M, Krzyzanowski W. Treatment of Achilles tendinopathy with platelet-rich plasma. *Int J Sports Med*. 2010;31(8):577-583.
26. Guillodo Y, Madouas G, Simon T, Le Dauphin H, Saraux A. Platelet-rich plasma (PRP) treatment of sports-related severe acute hamstring injuries. *Muscles Ligaments Tendons J*. 2016;5(4):284-288.
27. Haidich AB. Meta-analysis in medical research. *Hippokratia*. 2010;14(suppl 1):29-37.
28. Hall MP, Band PA, Meislin RJ, Jazrawi LM, Cardone DA. Platelet-rich plasma: current concepts and application in sports medicine. *J Am Acad Orthop Surg*. 2009;17(10):602-608.
29. Hamilton B, Tol JL, Almusa E, et al. Platelet-rich plasma does not enhance return to play in hamstring injuries: a randomised controlled trial. *Br J Sports Med*. 2015;49(14):943-950.
30. Harrison BC, Robinson D, Davison BJ, Foley B, Seda E, Byrnes WC. Treatment of exercise-induced muscle injury via hyperbaric oxygen therapy. *Med Sci Sports Exerc*. 2001;33(1):36-42.
31. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327(7414):557-560.
32. Howick J, Chalmers I, Glasziou P, et al; Oxford Centre for Evidence-Based Medicine. The Oxford 2011 levels of evidence. http://www.cebm.net/index.aspx?o=5653. Published 2011. Accessed September 14, 2018.
33. Hurley ET, Lim Fat D, Moran CJ, Mullett H. The efficacy of platelet-rich plasma and platelet-rich fibrin in arthroscopic rotator cuff repair: a meta-analysis of randomized controlled trials. *Am J Sports Med*. 2019;47(3):753-761.
34. Jakobsen RB, Engebretsen L, Slauterbeck JR. An analysis of the quality of cartilage repair studies. *J Bone Joint Surg Am*. 2005;87(10):2232-2239.
35. Joshi AB, Gill GS, Smith PL. Outcome in patients lost to follow-up. *J Arthroplasty*. 2003;18(2):149-153.
36. Komatsu DE, King L, Gurevich M, Kahn B, Paci JM. The in vivo impact of leukocyte injections on normal rat achilles tendons: potential detriment to tendon morphology, cellularity, and vascularity. *Am J Orthop (Belle Mead NJ)*. 2018;47(10). doi:10.12788/ajo.2018.0085
37. Kon E, Filardo G, Di Martino A, Marcacci M. Platelet-rich plasma (PRP) to treat sports injuries: evidence to support its use. *Knee Surg Sports Traumatol Arthrosc*. 2011;19(4):516-527.
38. Kujala UM, Orava S, Järvinen M. Hamstring injuries: current trends in treatment and prevention. *Sports Med*. 1997;23(6):397-404.
39. Levy GM, Lucas P, Hope N. Efficacy of a platelet-rich plasma injection for the treatment of proximal hamstring tendinopathy: a pilot study. *J Sci Med Sport*. 2019;22(3):247-252.
40. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *Ann Intern Med*. 2009;151(4):W65-W94.
41. Manduca ML, Straub SJ. Effectiveness of PRP injection in reducing recovery time of acute hamstring injury: a critically appraised topic. *J Sport Rehabil*. 2018;27(5):480-484.
42. Mason DL, Dickens V, Vail A. Rehabilitation for hamstring injuries. *Cochrane Database Syst Rev*. 2007;1:CD004575.
43. Mautner K, Colberg RE, Malanga G, et al. Outcomes after ultrasound-guided platelet-rich plasma injections for chronic tendinopathy: a multicenter, retrospective review. *PM R*. 2013;5(3):169-175.
44. Middleton KK, Barro V, Muller B, Terada S, Fu FH. Evaluation of the effects of platelet-rich plasma (PRP) therapy involved in the healing of sports-related soft tissue injuries. *Iowa Orthop J*. 2012;32:150-163.
45. Nguyen RT, Borg-Stein J, McInnis K. Applications of platelet-rich plasma in musculoskeletal and sports medicine: an evidence-based approach. *PM R*. 2011;3(3):226-250.
46. Ranganathan P, Pramesh CS, Aggarwal R. Common pitfalls in statistical analysis: intention-to-treat versus per-protocol analysis. *Perspect Clin Res*. 2016;7(3):144-146.
47. Rettig AC, Meyer S, Bhadra AK. Platelet-rich plasma in addition to rehabilitation for acute hamstring injuries in NFL players: clinical effects and time to return to play. *Orthop J Sports Med*. 2013;1(1):2325967113494354.

48. Reurink G, Goudswaard GJ, Moen MH, et al. Rationale, secondary outcome scores and 1-year follow-up of a randomised trial of platelet-rich plasma injections in acute hamstring muscle injury: the Dutch Hamstring Injection Therapy study. *Br J Sports Med*. 2015; 49(18):1206-1212.
49. Sánchez M, Anitua E, Andia I. Poor standardization in platelet-rich therapies hampers advancement. *Arthroscopy*. 2010;26(6):725-726.
50. Schippinger G, Prüller F, Divjak M, et al. Autologous platelet-rich plasma preparations: influence of nonsteroidal anti-inflammatory drugs on platelet function. *Orthop J Sports Med*. 2015;3(6): 2325967115588896.
51. Sheth U, Dwyer T, Smith I, et al. Does platelet-rich plasma lead to earlier return to sport when compared with conservative treatment in acute muscle injuries? A systematic review and meta-analysis. *Arthroscopy*. 2018;34(1):281-288.
52. Taylor LJ, Nabozny MJ, Steffens NM. A framework to improve surgeon communication in high-stakes surgical decisions: best case/ worst case. *JAMA Surg*. 2017;152(6):531-538.
53. Truss NJ, Armstrong PC, Liverani E, Vojnovic I, Warner TD. Heparin but not citrate anticoagulation of blood preserves platelet function for prolonged periods. *J Thromb Haemost*. 2009;7(11):1897-1905.
54. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol*. 2014;14:135.
55. Wetzel RJ, Patel RM, Terry MA. Platelet-rich plasma as an effective treatment for proximal hamstring injuries. *Orthopedics*. 2013;36(1):e64-e70.
56. Wright JG, Einhorn TA, Heckman JD. Grades of recommendation. *J Bone Joint Surg Am*. 2005;87(9):1909-1910.
57. Yerlikaya M, Talay Çaliş H, Tomruk Sütbeyaz S, et al. Comparison of effects of leukocyte-rich and leukocyte-poor platelet-rich plasma on pain and functionality in patients with lateral epicondylitis. *Arch Rheumatol*. 2017;33(1):73-79.
58. Yoshida R, Murray MM. Peripheral blood mononuclear cells enhance the anabolic effects of platelet-rich plasma on anterior cruciate ligament fibroblasts. *J Orthop Res*. 2013;31(1):29-34.
59. Zanon G, Combi F, Combi A, Perticarini L, Sammarchi L, Benazzo F. Platelet-rich plasma in the treatment of acute hamstring injuries in professional football players. *Joints*. 2016;4(1):17-23.
60. Zhang J, Wang JH. Platelet-rich plasma releasate promotes differentiation of tendon stem cells into active tenocytes. *Am J Sports Med*. 2010;38(12):2477-2486.
61. Zhou Y, Zhang J, Wu H, Hogan MV, Wang JH. The differential effects of leukocyte-containing and pure platelet-rich plasma (PRP) on tendon stem/progenitor cells—implications of PRP application for the clinical treatment of tendon injuries. *Stem Cell Res Ther*. 2015;6:173.

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